

BRANDON S. HAYES

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Campus Address: 1111 Engineering Drive • Boulder, CO 80303

Education

University of Colorado at Boulder, Boulder, CO

May 2018 – August 2024

- *Ph.D. Mechanical Engineering*

Rochester Institute of Technology, Rochester, NY

Aug. 2013 – May 2018

- *B.S. Biomedical Engineering*
- *B.S. Microelectronics Engineering*
 - *Math Minor (GPA: 3.96)*

Graduate Coursework

Heat Transfer
Fluid Dynamics

Mathematical Methods
Classical Thermodynamics

Automated Mechanical Design
CFD Unstructured Grids

Printed Electronics
Optimization Techniques

Experience

University of Colorado Boulder

Boulder, CO

Postdoctoral Researcher (advisor: Nathalie Vriend)

2024 – present

Fundamental Studies of Environmental Suspensions and Granular Flows

- Quantitative photoelastic analysis (forces and stress tensor) of soft granular media. Fabrication of large-scale photoelastic immersion-based rheometer to characterize fabric and force network of dry and wet particle suspensions.
- Micro to macro-scale research. How do particle-particle interactions (dry/wet) give rise to bulk frictional properties?
- DEM & DEM-CFD simulations (OpenFOAM) of particle-laden flows accounting for particle-wall, particle-particle, and fluid-particle interactions.
- Developed the arbitrary-shaped photoelastic grain solver (APEGS) codebase for quantitative photoelastic analysis of any-shaped 2.5D photoelastic particles. First of its kind to extend conventional PEGS codebase from its limitations of circular disks to any-shaped particles for shape-based analysis of granular systems and flows.

Quantica GmbH

Berlin, Germany

Visiting Researcher (supported by NSF INTERN grant)

Summer 2024

Complex and Highly Viscous Fluid Droplet Ejection for Inkjet Multi-Material 3D Printing

- Rheological analysis of viscoelastic polymers; inkjet drop watching of highly viscous (100 cP) and viscoelastic fluids.
- CFD simulations (OpenFOAM) of viscoelastic drop ejection and filament stretching dynamics.

University of Colorado Boulder

Boulder, CO

PhD Research (advisors: Robert MacCurdy and Gregory Whiting)

2018 – 2024

Thermal Bubble-Driven Micro-Pumps: The Building Blocks to Bring Microfluidics to the Masses

- Novel inkjet 3D printing approach (liquid/solid co-printing) of multi-material, 3D micro/mesofluidic devices; channels with integration of multi-material valves for flexible microfluidics, characterization of printed fluidic actuators.
- Femtosecond micro-milling for thermal inkjet (TIJ) resistor fabrication and microfluidics, bubble dynamics of TIJ resistors, development of a custom stroboscopic high-speed imaging setup (20 Mfps) for visualization of bubble dynamics, development of a custom electrical setup for 0.1-100 μ s, 0-300 V, 0-5.6 A heating pulses.
- CFD simulations (OpenFOAM and FLOW-3D) of thermal bubble-driven micro-pumps accounting for multi-physics coupling of thermal bubble nucleation, phase change, cavitation, and multiphase flow.
- Biocompatibility and reagent stability with thermal bubble-driven micro-pumps, investigation of organic fouling on TIJ resistors.
- Fluid-structure interaction of thin, elastic membranes for ultra-fast micro-robotic actuation and droplet ejection (experimentation and OpenFOAM + preCICE simulations).

Los Alamos National Laboratory
R&D SULI Intern (mentor: Pulak Nath)

Los Alamos, NM
Summer 2018

Microfluidic Applications for Single Cell Analysis and Ultra-Low Field NMR

- Adhesive transfer tape rapid prototyping of inertial focusing microfluidic devices for improved single cell capture efficiency, high speed imaging, particle image velocimetry, and MATLAB particle tracking for particle train analysis.
- COMSOL modeling of pre-polarized permanent magnet array and hexacoil electromagnetic coils for ultra-low field NMR detection of chemical warfare agents, optimization of hexacoil for < 25 ppm field uniformity in sample interrogation region and size reduction to a 2 in sphere.

RIT Senior Design Project (HP Sponsored)

Rochester, NY
2017 – 2018

Project Lead

Pulsatory Mixing of Laminar Flow using Bubble-Driven Micro-Pumps

- Full microfabrication process development (CVD and PVD, wet and dry etch, lithography, thermal, doping) for functional TIJ resistor on silicon, low-cost packaging methods for fluidic sealing.
- Stroboscopic imaging of bubble dynamics under different boiling regimes (nucleate and meta-stable), particle tracking, characterization of laminar flow mixing via particles and dye.
- Successful experimental demonstration of bubble-driven micro-mixer using TIJ technology.

HP Inc.

Corvallis, OR
Summer 2016, 2017

R&D Intern

Microfluidics Futures Group

- Characterization and demonstration of a microfluidic switchboard with integrated inertial pumps, stroboscopic / high-speed bubble imaging, fluorescence particle tracking, experimental analysis of fluid routing and sorting and valving.
- Experimental work with droplet microfluidics.
- Mathematical modeling of 2D rectangular flow via Green's functions to verify uniform particle distribution assumption
- Characterization of inertial pumps via high-speed imaging (1 Mfps), development of particle tracking methodology to transition from manual particle tracking to automated particle tracking.

Thermal Analysis and Microfluidics Lab

Rochester, NY
2015 – 2018

Undergraduate Researcher (mentor: Satish Kandlikar)

Enhancements of Pool Boiling

- Visualization of fluid convection patterns via PIV due to micro-milled surface geometry modifications in pool boiling.
- Process control of micro-milled surface geometry using laser confocal imaging and image processing.

ON Semiconductor

Rochester, NY
Summer 2015

R&D Intern

Process Engineer

- Process characterization and design of experiment for PECVD silicon oxynitride deposition to predict the effect of gas flows, RF power, and wafer distance on film stress, thickness, uniformity, and refractive index.
- Certified a new PECVD wafer susceptor for yearly material savings of \$70,000.

Selected Awards

- 1st Place at APS DSOFTE Gallery of Soft Matter Competition, Video Submission (2026)
- 1st Place at APS DFD Graduate Research Poster Competition for Best Poster (2023)
- 1st Place at ASME IMECE NSF-Sponsored Research Poster Competition for Best Poster (2023)
- Outstanding Mentor Award for CU Undergraduate Research Program (2022)
- Beverly-Sears Graduate Research Grant (2020)
- NSF GRFP Fellowship (2019)
- 1st Place at ASME IMECE Fluid Division's Young Engineer Paper Contest (2018)
- SULI Internship Program at Los Alamos National Labs (2018)
- Outstanding Undergraduate Scholar Award (2017)
- Preliminary Nomination for Barry Goldwater Scholarship (2015)

Selected Grants/Projects

NSF INTERN Supplemental Grant: *Complex Fluid Droplet Ejection for Inkjet Multi-Material 3D Printing* (2024)

- Research internship at Quantica in Berlin, Germany for study of piezoelectric inkjet printing of viscoelastic and complex fluids
- CU only: \$33,915 award value

Beverly-Sears Graduate Research Grant: *Stroboscopic Bubble Imaging* (2020)

- Development of stroboscopic bubble imaging system for printed TIJ resistors
- CU only: \$1,000 award value

NSF Graduate Research Fellowship Program: *M-Spice: An Integration Infrastructure Bringing Microfluidics to the Masses* (2019)

- Development of a VLSI integration infrastructure using inertial pumps for lab-on-a-chip and life science applications
- CU only: \$138,000 award value

RIT Senior Design Project: *Pulsatory Mixing of Laminar Flow using Bubble-Driven Micro-Pumps* (2017 – 2018)

- Development and experimental demonstration of a bubble-driven micro-mixer utilizing TIJ technology
- RIT and HP Inc. collaboration: \$5,000 project value

Teaching Experience

University of Colorado Boulder: master's and PhD student support. 3 total: 1 master's, 2 PhD. (2024-present)

University of Colorado Boulder: undergraduate research student supervision. 12 total, 1-2 each summer. (2019-present)

University of Colorado Boulder: teaching assistant for fluid dynamics (MCEN 3021). Project based undergraduate fluids course of 100-150 students. (2018-2019)

Journal Publications

Publications in Progress

1. Hayes, B., Kabutza, H., Patinoa, N., Jayaram, K., Ruzzene, M., and MacCurdy, R. Membrane-Assisted Thermal Inkjet (MA-TIJ) Droplet Ejection for TIJ-Incompatible Inks. TBD, 2026
2. Hayes, B., Bates, A., Chaudhuria, K., and Vriend, N. Arbitrary-Shaped Photoelastic Grain Solver (APEGS) for Non-Cylindrical 2D Granular Particles. TBD, 2026
3. Hayes, B., Gans, R., Chaudhuria, K., McMillan, B., Hayes, A., Burkea, H., McNamara, S., Chalklenc, T., and Vriend, N. Rheological and Photoelastic Response of Hydrating Soft Granular Particles. TBD, 2026

Publications in Refereed Journals

1. Kauzya, J.B., Hayes, B., Hayes, A., Thompson, J.F., Bellerjeau, C., Evans, K., Osio-Nosgaard, J., Gavai, G., Dikshit, K., Bruns, C., MacCurdy, R., Street, R., and Whiting, G. Experimental Analysis of Direct Ink Writing of Viscous Inks in Low Gravity Environments using Parabolic Flights. *Acta Astronautica*, **2024**
2. Hayes, B., Murphy, C., Rubio, J.M., Solis, D., Jayaram, K., and MacCurdy, R. Characterization of Organic Fouling on Thermal Bubble-Driven Micro-Pumps. *Biofouling*, **2024**
3. Smith, L., Hayes, B., Ford, K., Smith, E., and MacCurdy, R. Tunable Metamaterials for Impact Mitigation. *Advanced Materials Technologies*, **2024**
4. Hayes, B., Jayaram, K., and MacCurdy, R. Scaling Analysis of Thermal Bubble-Driven Micro-Pumps from Micro-Scale to Meso-Scale. *International Journal of Multiphase Flow*, **2023**
5. Hayes, B., Whiting, G.L., MacCurdy, R. An OpenFOAM Framework to Model Thermal Bubble-Driven Micro-Pumps. *Physics of Fluids*, **2023**
6. Hayes, B., Smith, L., Kabutz, H., Hayes, A., Jayaram, K., and MacCurdy, R. Rapid Fabrication of Low-Cost Thermal Bubble-Driven Micro-Pumps. *Micromachines*, **2022**

7. Hayes, B., Hainsworth, T., and MacCurdy, R. Liquid-Solid Co-Printing of Multi-Material 3D Fluidic Devices via Material Jetting. *Additive Manufacturing*, **2022**
8. Hayes, B., Whiting, G.L., MacCurdy, R. Modeling of Contactless Bubble-Bubble Interactions in Microchannels with Integrated Inertial Pumps. *Physics of Fluids*, **2021**
9. Hayes, B., Govyadinov, A., and Kornilovitch, P. Microfluidic Switchboards with Integrated Inertial Pumps. *Microfluidics and Nanofluidics*, **2017**

Conference Publications

- Hayes, B., Hayes, A., Rolleston, M., Ferreira, A., and Krisher, J. Pulsatory Mixing of Laminar Flow Using Bubble-Driven Micro-Pumps. *ASME International Mechanical Engineering Congress and Exposition*, **2018**

Invited Talks

- Arbitrary-Shaped Photoelastic Grain Solver (APEGS): Role of Shape on Granular Mechanics, *Gordon Research Conference Soft Matter*, **2026**

Selected Presentations

- Arbitrary-Shaped Photoelastic Grain Solver (APEGS) for Non-Cylindrical Soft Granular Particles, *Gordon Research Conference Soft Matter*, **2025**
- Ultra-Fast Micro-Actuation using Thermal Bubble-Driven Micro-Pumps, *ASME International Mechanical Engineering Congress and Exposition*, **2023**
- Thermal Bubble-Driven Micro-Pumps: The Building Blocks to Bring Microfluidics to the Masses, *ASME International Mechanical Engineering Congress and Exposition*, **2022**
- Liquid-Solid Co-Printing of Multi-Material 3D Fluidic Devices via Material Jetting, *Material Science Symposium at CU Boulder*, **2022**
- Particle Image Velocimetry Analysis of Bubble Induced Macroconvection Mechanisms for Enhanced Pool Boiling Performance, *International Conference on Nanochannels, Microchannels, and Millichannels*, **2017**

Selected Workshops

- Kavli Institute for Theoretical Physics (KITP) Program on Soft Earth Geophysics. University of California Santa Barbara, USA. Hosted by Professors Sujit Datta, Doug Jerolmack, Nathalie Vriend, and Vashan Wright, **2026**
- NSF Short Course Program on Batteries and Nanotechnology. Aristotle University, Thessaloniki, Greece. Hosted by Professor Katerina Aifantis, **2022**

US Patents

- 63/683,643 – Ultra-Fast Thermal Bubble Driven Actuation for Robotics and Associated Methods
- 63/583,540 – Actuation Devices and Methods, Membrane-Assisted Thermal Inkjet Droplet Ejection for Complex Inks and Associated Methods